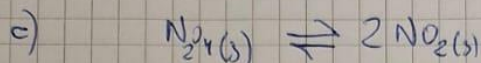


① a) Ec. recta $Y = -7255,5x + 18,962$

$\ln(K) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \Rightarrow -\frac{\Delta H^\circ}{R} = -7255,5$

$\Delta H^\circ = 60,3 \text{ kJ/mol}$

b) Endotermico



d) $0,01 \text{ M} \quad 0,05 \text{ M}$

$Q_c > K_c \Rightarrow$ se desplaza hacia reactivos

eq) $(0,01+x) \text{ M} \quad (0,05-2x) \text{ M}$

$[\text{N}_2\text{O}_4]_0 = 0,01 \text{ M} \quad [\text{NO}_2]_0 = 0,05 \text{ M}$

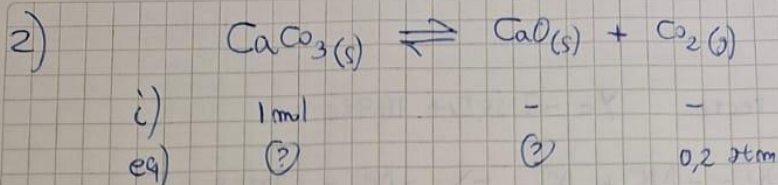
$K_c = \frac{([\text{NO}_2]_0 - 2x)^2}{([\text{N}_2\text{O}_4]_0 + x)} \Rightarrow 4x^2 - (K_c + 4[\text{NO}_2]_0)x + [\text{NO}_2]_0^2 - K_c[\text{N}_2\text{O}_4]_0 = 0$

$x = \frac{0,019}{0,032}$

$[\text{NO}_2]_{eq} = 0,012 \text{ M}$
 $[\text{N}_2\text{O}_4]_{eq} = 0,029 \text{ M}$

d) $K_p = K_c (RT)^{\Delta n} \Rightarrow K_p = K_c RT = 91129$

$\Delta G^\circ = -RT \ln(K_p) = 5,4 \text{ kJ/mol}$



a) $[\text{CO}_2]_{\text{eq}} = \frac{P}{RT} = 4,88 \cdot 10^{-3} \text{ M}$

b) $m_{\text{CaCO}_3} = 1 \text{ mol} - m_{\text{CO}_2} = 0,95 \text{ mol}$

$m_{\text{CaCO}_3} = m_{\text{CO}_2} = \frac{P \cdot V}{RT} = 0,049 \text{ mol}$

c) $K_p = P_{\text{CO}_2_{\text{eq}}} = 0,2$ $K_c = [\text{CO}_2]_{\text{eq}} = 4,88 \cdot 10^{-3}$

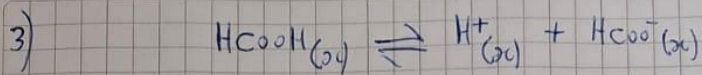
d) $\uparrow P \Rightarrow \uparrow Q (Q > K) \Rightarrow$ desplaza a reactivos

e) $\Delta H > 0 \Rightarrow \text{Si } \uparrow T \rightarrow$ desplaza a productos

f) $\downarrow m_{\text{CO}_2} \Rightarrow \downarrow Q (Q < K) \Rightarrow$ desplaza a productos

$\downarrow m_{\text{CaCO}_3}$ $\downarrow m_{\text{CaO}}$
 $\uparrow m_{\text{CaO}}$ $\downarrow m_{\text{CaCO}_3}$

g) $\uparrow m_{\text{CaCO}_3} \Rightarrow Q$ no cambia



c) $\underbrace{25 \text{ mL } 0,1 \text{ M}}_{2,5 \cdot 10^{-3} \text{ mol}}$

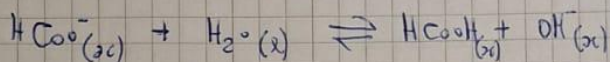
$\text{pK}_a = 3,75$

$\downarrow$
 $\text{K}_a = 10^{-3,75} = 1,78 \cdot 10^{-4}$

a) $m_{\text{KOH}} = m_{\text{HCOOH}}$ (Punto equivalencia)

$V_{\text{KOH}} = \frac{0,1 \text{ M} \cdot 99,2 \text{ mL}}{0,05 \text{ M}} = 0,05 \text{ L}$

b) $V = 25 \text{ mL} + 50 \text{ mL} = 75 \text{ mL}$ (Vol. aditivos)



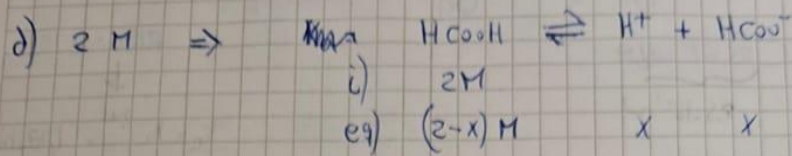
i) $\frac{2,5 \cdot 10^{-3} \text{ mol}}{\text{eq } 75 \text{ mL}}$
 $(0,033 \text{ M})$

eq) $(0,033 - x) \text{ M}$

$x \quad x$

$K_b = \frac{K_w}{K_a} = \frac{5,62 \cdot 10^{-11}}{0,033 - x} = \frac{x^2}{0,033 - x} \Rightarrow x = \sqrt{\frac{K_b \cdot 0,033}{1}} = 1,36 \cdot 10^{-6} \text{ M}$
desprecia $\text{pH} = 8,13$

c) Añade $\text{HCOO}^- \Rightarrow \uparrow \alpha$ ($\alpha > k$) $\rightarrow$ desplace a productos
 $\uparrow [\text{HCOO}^-] \Rightarrow \uparrow \text{pH}$



$$K_a = \frac{x^2}{[\text{HCOOH}]_0 - x}$$

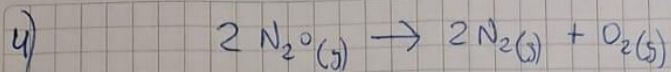
$\downarrow$
desprecia

$$x = \sqrt{K_a [\text{HCOOH}]_0} \Rightarrow \alpha = \frac{x}{[\text{HCOOH}]_0} = \sqrt{\frac{K_a}{[\text{HCOOH}]_0}}$$

$$[\text{HCOOH}]_0 = 2 \text{ M} \Rightarrow \alpha = 0,94 \%$$

$[\text{HCOOH}]_0 = 0,1 \text{ M} \rightarrow$ Es como agregar H_2O
 $\hookrightarrow$ desplace a productos (Le Chatellier)
 $\hookrightarrow \uparrow \alpha$

$\alpha = 13,3 \%$



$k = 3,5 \cdot 10^{-3} \frac{1}{s}$
 order 1

a) $v = k [\text{N}_2\text{O}]$

b) $\ln [\text{N}_2\text{O}]_t = -2kt + \ln [\text{N}_2\text{O}]_0$

$\ln [\text{N}_2\text{O}]_t = -2 \cdot 3,5 \cdot 10^{-3} \frac{1}{s} \cdot 200 s + \ln(9,5)$

$[\text{N}_2\text{O}]_{200s} = 0,12 \text{ M}$

c) $t_{1/2} = \frac{\ln(2)}{2k} = 99 \text{ s}$

d) $\ln\left(\frac{k_1}{k_2}\right) = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right) \Rightarrow E_a = 71,24 \text{ kJ/mol}$
 $\frac{k_1}{k_2} = \frac{3,5 \cdot 10^{-3} \text{ 1/s}}{1,05 \cdot 10^{-2} \text{ 1/s}}$
 $\frac{1}{T_1} = \frac{1}{600 \text{ K}}$
 $\frac{1}{T_2} = \frac{1}{650 \text{ K}}$